

1 of 10

The system call that is used to load and run a program is _____

- | | |
|---|-------------------------------|
| ☐ A exec | ☐ B execv |
| ☐ C fork | ☐ D exit |
| ☒ E A or B | |

SUBMIT ANSWER

2 of 10

The system call in the previous question that requires that parameters for the program must be specified as an array is _____

- | | |
|--------------------------------|--|
| ☐ A exec | ☒ B execv |
| ☐ C fork | ☐ D exit |
| ☐ E A or B | |

SUBMIT ANSWER

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The system call that is used by a parent process to wait for its child process to finish its job is _____

- | | |
|---|-------------------------------|
| ☐ A exec | ☐ B execv |
| ☐ C exit | ☐ D fork |
| ☒ E wait | |

SUBMIT ANSWER

4 of 10

The system call that is used by a process to terminate its work and send a status value back to its parent process is _____

- | | |
|---|-------------------------------|
| ☐ A exec | ☐ B execv |
| ☒ C exit | ☐ D fork |
| ☐ E wait | |

SUBMIT ANSWER

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a Unix command that is used to display a process status is _____

- | | |
|---------------------------------------|--------------------------------|
| ☒ A ps | ☐ B getpid |
| ☐ C process | ☐ D status |
| ☐ E man | |

SUBMIT ANSWER

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wait(NULL); means a parent process will not wait for its child process.

T True

F False

SUBMIT ANSWER

7 of 10

```

Considering the following C program;
/* forkAndExec.c */
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
int main()
{
    pid_t pid;
    int i;
    for (i=0; i<3; i++) {
        pid = fork();
        if (pid == 0) {
            if (i == 0) {
                execlp("echo", "echo", "good morning");
                printf("OK\n");
            }
            else if (i == 1) {
                execlp("echo", "echo", "good afternoon");
                printf("OK\n");
            }
            else {
                execlp("echo", "echo", "good night");
                printf("OK\n");
            }
        }
    }
    while(wait(NULL) != -1);
    printf("goodbye\n");
    return 0;
}

```

How many lines of output will be printed out?

A 1
B 2
C 3
D 4
E 5

A 1
B 2
C 3
D 4
E 5

4

10 of 10

What is (are) the possible outputs of the program?

A good morning
good afternoon
good night
goodbye

B good morning
goodbye
good afternoon
good night

A good afternoon
good night
good morning
goodbye

B goodbye
good morning
good afternoon
good night

C All of the above.

SUBMIT ANSWER

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```

Consider the following C program;
/* forkAndExec.c */
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
int main()
{
    pid_t pid;
    int i;
    for (i=0; i<3; i++) {
        pid = fork();
        if (pid == 0) {
            if (i == 0) {
                execlp("cd", "cd", "test");
                printf("OK\n");
            }
            else if (i == 1) {
                execlp("echo", "echo", "good afternoon");
                printf("OK\n");
            }
            else {
                execlp("echo", "echo", "good night");
                printf("OK\n");
            }
        }
    }
    while(wait(NULL) != -1);
    printf("goodbye\n");
    return 0;
}

```

First child failed to exit cd

How many lines of output will be printed out?

A 4
B 5
C 6
D 7
E 8

A 4
B 5
C 6
D 7
E 8

7

9 of 10

How many times that the word "goodbye" will be printed out?

A 1
B 2
C 3
D 4
E 5

A 1
B 2
C 3
D 4
E 5

2

SUBMIT ANSWER

Week 6

1 of 7

A thread uses program counter of the process that created it.

☒ True ☐ False

SUBMIT ANSWER

Thread has its own separate Program Counter, so False

2 of 7

Threads created from the same process may have different priorities.

☒ True ☐ False

SUBMIT ANSWER

3 of 7

What kind of OS is the JVM?

☐ A single-process single-threaded ☒ B single-process multi-threaded
☐ C multi-process single-threaded ☐ D multi-process multi-threaded
☐ E B was D

SUBMIT ANSWER

4 of 7

What kind of OS of the Windows 10?

☐ A single-process single-threaded ☐ B single-process multi-threaded
☐ C multi-process single-threaded ☒ D multi-process multi-threaded
☐ E B was D

SUBMIT ANSWER

5 of 7

What kind(s) of mapping between User-thread and kernel-thread that allow(s) more than one user thread to be executed at a time.

☐ A many-to-one ☐ B many-to-many
☐ C one-to-one ☐ D A was B
☒ E B was C

SUBMIT ANSWER

6 of 7

Each thread does need a data structure for its own information since it uses the data structure from its process's PCB.

☒ True ☐ False

SUBMIT ANSWER

Each thread must have its own thread control block location, but this can be pointed to Process Control Block

7 of 7

If you use a thread library, the benefit is your program will be run faster.

☐ True ☒ False

SUBMIT ANSWER

Real benefit is source code portable

1 of 16

For Pthread library, the function that we need a thread to execute must be named runner.

☒ True ☐ False

SUBMIT ANSWER

No name needed for thread

2 of 16

The Pthread library function that works similar to the wait function of a parent process is ____.

☐ A pthread_attr_t
☐ B pthread_create()
☒ C pthread_join()
☐ D pthread_wait()
☐ E pthread_exit()

SUBMIT ANSWER

3 of 16

If you write a C program with the main function as follows: main(int argc, char *argv[]). After compiling the source code assumes that you have the program named prog. If you call the program using the following command: prog 1 2 3. What is the value of the argc?

☐ A 1
☒ B 2
☐ C 3
☐ D 4
☐ E 5

SUBMIT ANSWER

4, since program name is also counted

4 of 16

From the previous question, what is the value of argv[1]?

☐ A 1
☒ B "1"
☐ C 2
☐ D "2"
☐ E "prog1"

SUBMIT ANSWER

Return as string na ja

5 of 16

If you write the Java program named Prog1 with the following main method: main(String[] args). Assuming that you run the program using the following command: java Prog1 1 2 3. What is the value of args.length?

☐ A 1
☒ B 2
☐ C 3
☐ D 4
☐ E 5

SUBMIT ANSWER

3, since Java only starts counting the parameter; class name is not counted

6 of 16

From the previous question, what is the value of args[1]?

☐ A 1
☒ B "1"
☐ C 2
☐ D "2"
☐ E "Prog1"

SUBMIT ANSWER

"2", since Java only starts counting the parameter; class name is not counted

7 of 16

Assuming you have the C source code file named test.c. Assuming this code uses Pthread library. In order to compile the code to get the program named test, What is (are) the command(s) that you can use?

☐ A gcc test.c
☐ B gcc -o test test.c
☐ C gcc test.c -l pthread
☒ D gcc -o test test.c -l pthread
☐ E C vla D

SUBMIT ANSWER

C would result in a.out only

8 of 16

In order to create a Java class that can be executed as a separate thread, that class needs to inherit from Java ____ class.

☒ A Runnable
☐ B thread
☐ C Pthread
☐ D Win32
☐ E A vla B

SUBMIT ANSWER

Runnable is interface and not class, needed to be implement

9 of 16

If we do not want to use the class in the previous question, the Java class that we need it to be executed as a separate thread will implement the Java _____ interface.

- | | |
|---|--------------------------------|
| ☒ A Runnable | ☐ B Thread |
| ☐ C Pthread | ☐ D Win32 |
| ☐ E A via B | |

SUBMIT ANSWER

10 of 16

The method that can be considered the main method of a class that will be executed as a separate Thread is _____

- | | |
|--|--------------------------------|
| ☒ A run | ☐ B runner |
| ☐ C start | ☐ D join |
| ☐ E wait | |

SUBMIT ANSWER

11 of 16

Java thread method that is similar to wait() in the concept of process is _____

- | | |
|-------------------------------|---|
| ☐ A run | ☐ B runner |
| ☐ C start | ☒ D join |
| ☐ E wait | |

SUBMIT ANSWER

14 of 16

A thread terminates immediately after it was interrupted is the idea of _____

- | | |
|--|---|
| ☒ A Asynchronous Cancellation | ☐ B Deferred Cancellation |
| ☐ C Synchronous Cancellation | ☒ D A via B |
| ☐ E B via C | |

Deferred means the thread can run for the safe place
In Async, the thread must stop immediately

SUBMIT ANSWER

15 of 16

The idea of creates one or more threads in advanced so that it can be executed immediately when it is needed is called _____.

- | | |
|--|--|
| ☐ A Thread Cancellation | ☒ B Thread Pool |
| ☐ C Thread Synchronization | ☐ D Thread Interrupted |
| ☐ E Thread Scheduling | |

SUBMIT ANSWER

16 of 16

```
From the following C program
/*pthread.c*/
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
void *runner(void *param);
int main(int argc, char *argv[]) {
    pthread_t tid;
    pthread_attr_t attr;
    int sum;
    if (argc != 2) {
        fprintf(stderr, "usage: pthread <integer value>\n");
        exit(1);
    }
    if (atoi(argv[1]) < 0) {
        fprintf(stderr, "number must be >= 0\n");
        exit(2);
    }
    pthread_attr_t attr;
    pthread_create(&tid, &attr, runner, argv[1]);
    printf("I am mother thread, I will wait for my child thread\n");
    pthread_join(tid, NULL);
    printf("from my child sum = %d\n", sum);
}
void *runner (void *param) {
    int i;
    int upper = atoi(param);
    int sum = 0;
    printf("I am child thread, I am calculating\n");
    if (upper > 0) {
        for (i = 0; i <= upper; ++i) {
            sum += i;
        }
    }
    pthread_exit(0);
}
```

If a user runs this program using the following command ./pthread 5. What is (are) the possible output[s]?

- | | |
|--|--|
| ☐ A I am mother thread, I will wait for my child thread
I am child thread, I am calculating
from my child sum = 0 | ☐ B I am mother thread, I will wait for my child thread
I am child thread, I am calculating
from my child sum = 15 |
| ☒ C I am child thread, I am calculating
I am mother thread, I will wait for my child thread
from my child sum = 0 | ☐ D I am mother thread, I will wait for my child thread
I am child thread, I am calculating
from my child sum = 15 |
| ☐ E no output | |

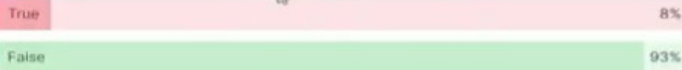
pthread_exit(0) -> the 'sum' getting from thread is 0, rather than the actual sum
Both mother and Child can happen since it is located before join

SUBMIT ANSWER

10. Pthread supports only Deferren Cancellation.

Hide Results

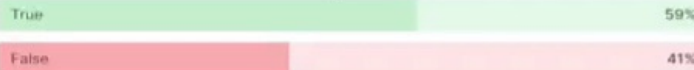
40/43 Students Answered



11. sleep(1); is considered a cancellation poubt in Pthread

Hide Results

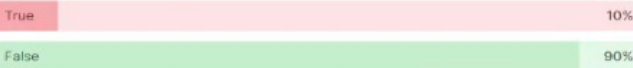
40/43 Students Answered



12. The maximum number of threads in the Java Thread Pool is 10.

Hide Results

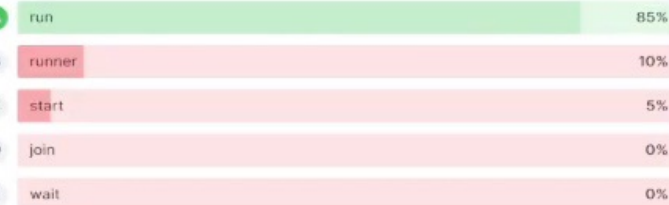
40/43 Students Answered



13. The method that can be considered the main method of a class that will be executed as a separate Thread is _____.

Hide Results

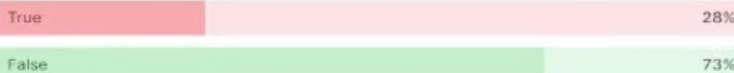
40/43 Students Answered



15. Creating an object of a class that inherits from the Java Thread class means a new fully operable thread is already created.

Hide Results

40/43 Students Answered



16. When the method start() is called a Java thread is in the running state.

Hide Results

40/43 Students Answered



20. From the Java program in the picture.

If a user runs this program using the following command

java Main 6

What is the possible output?

Hide Results 40/43 Students Answered

A	I am calculating 15 Goodbye	5%
B	I am calculating 0 Goodbye	10%
☒ C	I am calculating 21 Goodbye	80%
D	I am calculating Goodbye 15	3%
E	I am calculating Goodbye 0	0%
F	I am calculating Goodbye 21	3%

```
1 class SharedData {
2     private int sum;
3     public int getSum() {
4         return sum;
5     }
6     public void calculateSum(int num) {
7         sum = num * (num+1) / 2;
8     }
9 }
10 class Child extends Thread {
11     private SharedData data;
12     int num;
13     public Child(SharedData data, int num) {
14         this.data = data;
15         this.num = num;
16     }
17     public void run() {
18         System.out.println("I am calculating");
19         data.calculateSum(num);
20     }
21 }
22 public class Main {
23     public static void main(String[] args) {
24         int num = Integer.parseInt(args[0]);
25         SharedData data = new SharedData();
26         Child child = new Child(data, num);
27         child.start();
28         try {
29             child.join();
30         }
31         catch (InterruptedException ie) {}
32         System.out.println(data.getSum());
33         System.out.println("Goodbye");
34     }
35 }
36 }
```

Week 7

1 of 10

OS Scheduler is responsible for context switching.

☒ True ☐ False

SUBMIT ANSWER

Context Switch is Dispatcher job, Scheduler is the one who choose

2 of 10

Round robin scheduling gives good response time.

☒ True ☐ False

SUBMIT ANSWER

3 of 10

For multi-level feedback queue, a process always moves from a higher priority queue to a lower priority queue.

☐ True ☒ False

SUBMIT ANSWER

ML Queue is Fixed-Priority Queue -> Possibility of Starvation
ML Feedback Queue can change the priority

4 of 10

Aging can be used to solve process starvation problem.

☒ True ☐ False

SUBMIT ANSWER

5 of 10

If the waiting time is increased, the turnaround time is also increased.

☒ True ☐ False

SUBMIT ANSWER

Waiting Time = Turnaround Time – Burst Time

6 of 10

Improvement of response time makes better turnaround time.

☒ True ☐ False

SUBMIT ANSWER

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From the following information:

Process	Arrival time	Burst Time
P1	0	8
P2	2	4
P3	1	1

Create Gantt Chart First:

P1 -> P3 -> P2

P3 comes first since P3 has the shortest job

Turnaround Time (TT) = (Wait) + Burst - Arrival Time

What is the average turn around time? If the scheduling algorithm is SJF.

P1 TT: (0)+8-0 = 8

P2 TT: (8+1)+4-2 = 11

P3 TT: (8)+1-1 = 8

Avg. TT -> (8+11+8)/3 = 9

SUBMIT ANSWER

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From the following information;

Process	Arrival time	Burst Time
P1	0	8
P2	2	4
P3	1	1

Create Gantt Chart First:

P1 -> P3 -> P2

P3 comes first since P3 has the shortest job

Waiting Time = Turnaround Time - Burst Time

What is the average waiting time? If the scheduling algorithm is SJF.

Enter Answer Here

P1 WT: 8-8 = 0

P2 WT: 11-4 = 7

P3 WT: 8-1 = 7

$14/3 = 4.66$
Avg. WT -> (0+7+7)/3 = ~~4.33~~

SUBMIT ANSWER

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From the following information;

Process	Arrival time	Burst Time
P1	0	8
P2	2	4
P3	1	1

Create Gantt Chart First:

P1 -> P3 -> P2 -> P1

SRTF is preemptive -> can change queue

Turnaround Time (TT) = (Wait) + Burst - Arrival Time

What is the average turn around time? If the scheduling algorithm is SRTF.

P1 TT: (0+1'+4')+(1+7')+0 = 13

P2 TT: (1+1)+4-2 = 4

P3 TT: (1)+1-1 = 1

Avg. TT -> (13+4+1)/3 = 6

SUBMIT ANSWER

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From the following information;

Process	Arrival time	Burst Time
P1	0	8
P2	2	4
P3	1	1

Create Gantt Chart First:

P1 -> P3 -> P2 -> P1

SRTF is preemptive -> can change queue

Waiting Time = Turnaround Time - Burst Time

What is the average waiting time? If the scheduling algorithm is SRTF.

P1 WT: 13-8 = 5

P2 WT: 4-4 = 0

P3 WT: 1-1 = 0

Avg. WT -> (5+0+0)/3 = ~~1.33~~
 $5/3 = 1.667$

SUBMIT ANSWER

11. From the following information;
 Process Arrival time Burst Time
 P1 0 8
 P2 2 4
 P3 11

What is the turnaround time of P1, if the algorithm is Round Robin with time quantum 4.

Ans = 13

Hide Answers

Show Names

37/43 Students Answered

7	👁
7	👁
10	👁
13	👁
13	👁
13	👁

P1	P3	P2	P1
0	4	5	9
			13

$$TP1 = 13 - 0 = 13$$

$$TP2 = 9 - 2 = 7$$

$$TP3 = 5 - 1 = 4$$

$$WT1 = 13 - 8 = 5$$

$$WT2 = 7 - 4 = 3$$

$$WT3 = 4 - 1 = 3$$

12. From the following information;
 Process Arrival time Burst Time
 P1 0 8
 P2 2 4
 P3 11

What is the turnaround time of P2, if the algorithm is Round Robin with time quantum 4.

Ans = 7

Hide Answers

Show Names

37/43 Students Answered

9	👁
6	👁
2	👁
7	👁
7	👁

13. From the following information;
 Process Arrival time Burst Time
 P1 0 8
 P2 2 4
 P3 11

What is the turnaround time of P3, if the algorithm is Round Robin with time quantum 4.

Ans = 4

Hide Answers

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37/43 Students Answered

4	👁
2	👁
8	👁